



MINING DEALER BEST PRACTICE

Mobile Hangar for Welding Truck Bodies in the Field

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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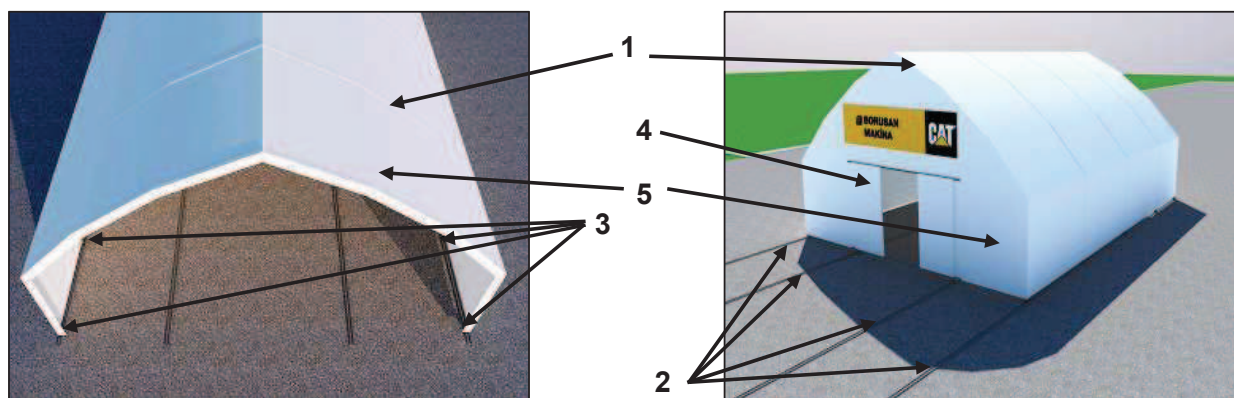
1.0 Introduction

For high quality welding of truck bodies, it is important to maintain correct temperature of the welding area. As per the CAT requirements, for quality seams, the temperature of the metal in the welding area should be maintained at certain temperatures. If welding needs to be performed in the field, these conditions can be challenging to provide, especially in severe winter conditions.

In our two projects (Bozshakol & Aktogay), the welding operations on truck bodies occurred in winter. Besides, there was no possibility to weld bodies in shops, because the customers did not have any at that time. We had expected it. Therefore, we developed the mobile hangar structure that will close the welding area from ambient weather conditions and enable us to keep the correct welding temperature inside the hangar.

2.0 Best Practice Description

The mobile hangar consists of (1) Framework, (2) Guiding rails, (3) Rollers, (4) Sliding Doors with a Guide and (5) Removable Curtain (covering material of the hangar).



Truck body assembling procedure usually consists of three main stages:

1. Joining main parts of the body by tack welding;
2. Welding seams on the one side;
3. Turning over the body and welding the seams on the other side.

Our hangar was designed for welding bodies of 785C and 777D/F trucks in the field. We were striving to make the dimensions of the hangar as small as possible in order not to heat the extra volume inside a hangar. The main problem was turning over the body. Obviously, it was impossible to place the crane inside the hangar. That is why we did the following:

- 1) The hangar was assembled by the crane in the open air;
- 2) The hangar with the end wall removed was moved on the rails onto the joined body;

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- 3) The end wall was installed in the hangar. The hangar was heated and the body was welded on one side;
- 4) The end wall was removed and the hangar was moved on the rails back.
- 5) The body was turned over by the crane on the open air;
- 6) The hangar was moved on the rails again back onto the body turned over;
- 7) The end wall was installed in the hangar. The hangar was heated again and the body was welded on the other side;
- 8) The end wall was removed and the hangar was moved on the rails back and the body was ready!

Thanks to this hangar, in the conditions of the hard winter we have finished the welding of 35 x 785 Truck bodies as per the schedule and with a good quality.

Below several photos illustrating the implementation of this idea in the project.



The materials for the hangar have been delivered and placed in the welding area



The framework of the hangar is being assembled by the crane



The hangar has been assembled with no end wall installed



The body is inside the hangar and the end wall has been installed. The hangar has been heated and ready for use

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The welders are welding the body parts

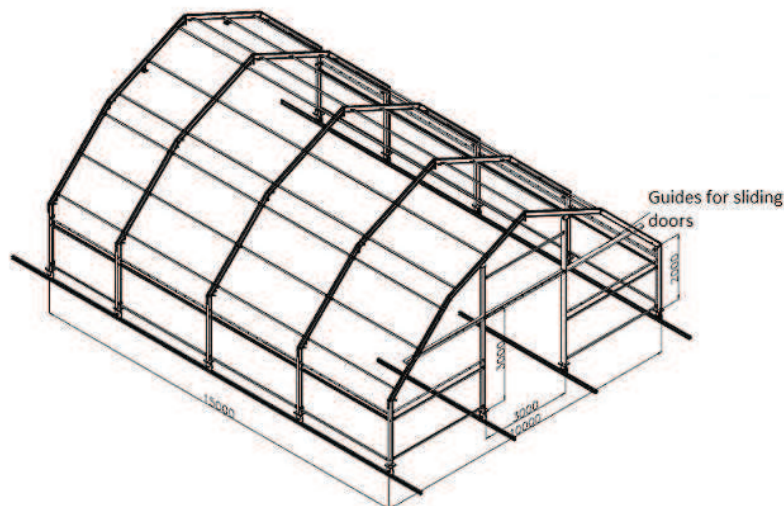


The welding of the body parts has been made on one side

3.0 Implementation Steps

- Decide what model of the truck body will be welded in the hangar;
- If 785C truck bodies, you can use dimensions of our hangar. For larger models, you should work with the construction department to determine dimensions;
- Purchase the materials needed;
- Provide the delivery of the materials for the hangar into the body welding area;
- Provide the crane and the people who will assemble/ dismantle the hangar in the welding spot.

Mobile Hangar Structure



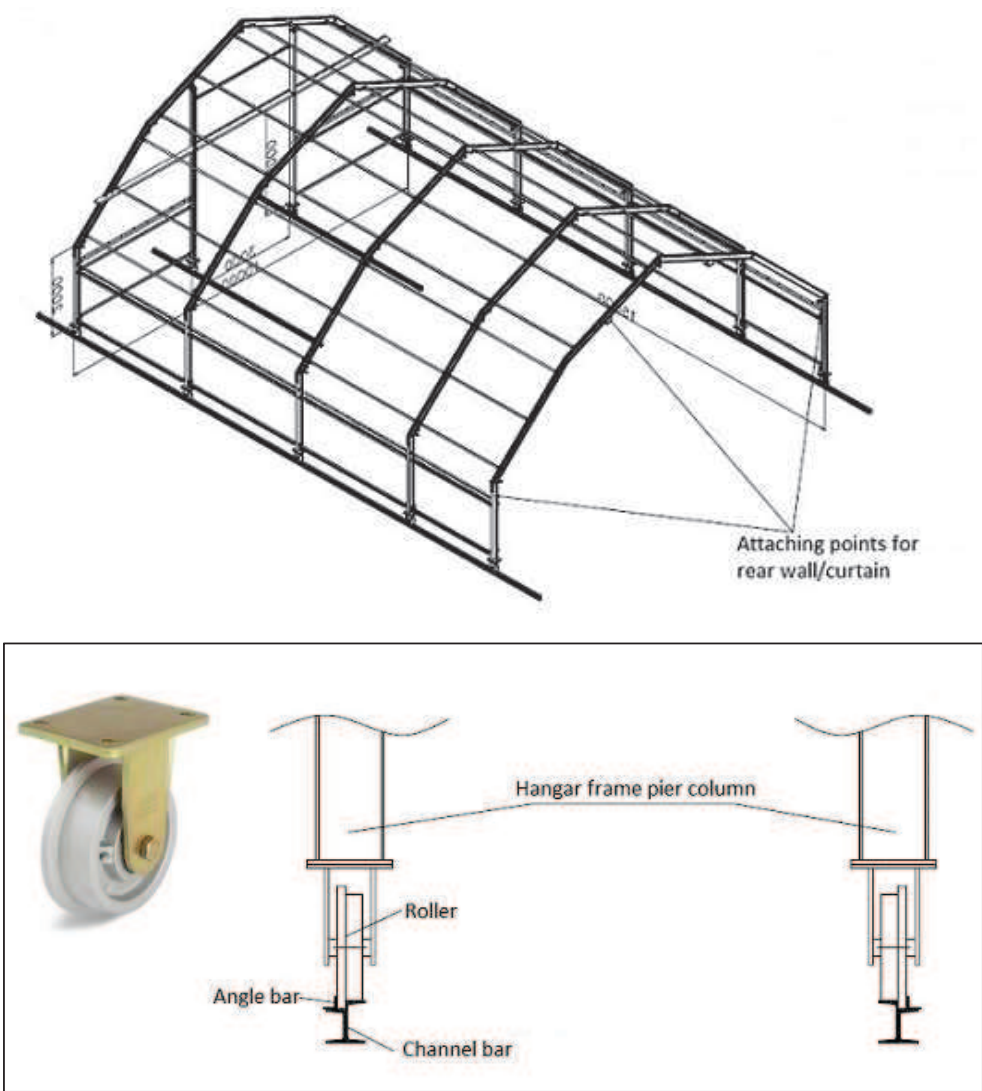
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4.0 Benefits

- It provides the convenient and required conditions for a good quality welding of truck bodies in the field, especially in the cold seasons of the year;
- Due to its mobile and light design, three people can easily move it during its use, because the whole structure rests on the rollers, which move along the guiding rails.
- It may save significant time and money required otherwise for transportation of truck bodies in pieces to closed welding shop and back.

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5.0 Resources Required

- Purchase a proper size hangar from supplier
- Crane to assemble/ dismantle the hangar.

6.0 Supporting Attachments / References

N/a.

7.0 Related Best Practices

N/a.

8.0 Acknowledgements

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